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SUPERSEDING
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FEDERAL SPECIFICATION

SLAG AND STONE, CRUSHED; GRAVEL, CRUSHED AND UNCRUSHED (FOR SEWAGE TRICKLING FILTER MEDIA)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers crushed slag and stone, crushed and uncrushed gravel, used in sewage trickling filters.

1.2 Classification. The material shall be of the following types, classes, and grades, as specified (see 6.2):

Type I—Stone, crushed.
Type II—Slag, crushed, blast-furnace.
Type III—Gravel.
  Class 1—Crushed.
  Class 2—Uncrushed.
Grade A—For freezing temperature.
Grade B—For above freezing temperature.

1.3 Sizes. The material shall be of the following nominal sizes, unless otherwise specified (see 6.2):

Inches:
3-1/2 to 2-1/2
3-1/2 to 2
3 to 2
3 to 1-1/2
2-1/2 to 1-1/2

2. APPLICABLE DOCUMENTS

2.1 Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply:

American Society For Testing And Materials (ASTM) Publications:

C 88—Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate.
C 136—Sieve or Screen Analysis of Fine or Coarse Aggregates.
D 75—Sampling Stone, Slag, Gravel, Sand, and Stone Block, for Use as Highway Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material.

3.1.1 *All types, classes, and grades.* The filter media shall be of sound, durable, and unweathered materials, suitable for service in the trickling filter treatment of sewage, and shall meet all tests in 4.4. The material shall be free from grease, oil, dirt, clay, sand and any other organic material undesirable in a filter media.

3.1.2 *Type II, crushed blast-furnace slag.* This material shall meet all other requirements of media, and in addition, shall be passed over by a magnetic separator of such size and strength that all free iron is removed.

3.2 Grades.

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3.2.1 *Grade A.* This grade of material shall meet all other requirements of media, and in addition, shall be resistant to deterioration when subjected to freezing and thawing temperatures when tested in accordance with 4.4.2.

3.2.2 *Grade B.* This grade of material shall meet all other requirements of media, except resistance to freezing and thawing temperatures is not required.

3.3 Grading sizes. Filter media shall be as uniform in size as practicable when tested in accordance with 4.4.1 and shall conform to grading shown in table I for the specified nominal size.

TABLE I. *Grading sizes*

Nominal size square openings inches	Amount passing sieve (square openings) percent by weight						
	4 in.	3-1/2 in.	3 in.	2-1/2 in.	2 in.	1-1/2 in.	1 in.
3-1/2 to 2-1/2	100	90-100	—	0-10	0-2	—	—
3-1/2 to 2	100	90-100	—	—	0-10	0-2	—
3 to 2	100	100	90-100	—	0-10	0-2	—
3 to 1-1/2	100	100	90-100	—	—	0-10	0-2
2-1/2 to 1-1/2	100	100	100	90-100	—	0-10	0-2

3.4 Particle shape. The individual pieces of the material shall be as nearly equidimensional as practicable. Pieces that are elongated or flat, or both, shall not exceed 10 percent by weight. These shapes are defined as having length exceeding width, and width exceeding thickness in ratio of more than 1.5 to 1.0, dimensions being maximum for the size of pieces under consideration.

3.5 Soundness. Media shall not disintegrate by flaking, softening, splitting, or spalling when subjected to periodic wetting and drying, and the variation in temperature occuring in a trickling filter. When subjected to 20 cycles, the loss shall not exceed 10 percent, and the combined loss shall not exceed 15 percent by weight, due to breaking, crumbling, or flaking when tested in accordance with 4.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *Lot.* Unless otherwise specified (see 6.2), a lot shall consist of 250 tons of material or fraction thereof of each type, class, grade, or size offered for delivery at one time.

4.2.2 *Visual examination and tests.* Unless otherwise specified (see 4.4.2), for soundness test, samples for examination and tests shall be in accordance with ASTM D 75 and one sample shall be taken from each lot.

4.3 Visual examination. Examination of the end item shall be in accordance with the following defects:

(a) Type, class, or grade not as specified.
(b) Oil, dirt, clay, or other extraneous matter present.
(c) Elongated or flat pieces, or both, exceeding 10 percent of the sample by weight.
(d) Not free of soft or disintegrated pieces.
(e) Particle shape not equidimensional.

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4.4 Tests.

4.4.1 *Grading, sieve analysis; all types, classes, and grades.* Tests shall be conducted in accordance with ASTM C 136, using a sieve with square openings (see 3.3).

4.4.2 *Soundness, grade A.* Tests for soundness shall be conducted in accordance with the test specified in ASTM C 88, except as modified by the following:

4.4.2.1 *Preparation of test sample.* The test sample shall consist of at least 20 pieces of the size specified for use in the filter. This sample shall be prepared in the testing laboratory either by crushing, if the sample furnished is one of ledge rock or boulders, or by selecting pieces from the sample furnished from stock material. The test sample shall be prepared so as to be representative in respect to variations in color, texture, density, porosity, and geological classification. The test sample shall be screened and the weight retained on each sieve recorded. Each piece shall be individually weighed, identified, and placed either in an individual wire mesh container or in a wire mesh basket provided with suitable separators so that the identity of each test piece will be retained throughout the test.

4.4.2.2 *Continuity of test.* If for any reason the continuity of the test is interrupted, such as over a weekend, the sample shall be kept in a dry condition at room temperature.

4.4.2.3 *Qualitative examination.* At the end of each cycle, each piece shall be examined for signs of disintegration, such as cracking, splitting, checking, flaking, or crumbling. The portions that can be removed with the fingers shall be taken off.

4.4.2.4 *Quantitative examination.* At the end of the twentieth cycle, or an earlier cycle if the sample has obviously failed to pass the test, the percentage loss shall be determined by the three following methods:

(a) *Loss by breaking.* An individual piece which has broken or split into three or more fragments or which has so cracked that subsequent breakage into three pieces is beyond question, shall be considered to have failed the test, provided each of three such fragments is at least 10 percent of the original piece by weight. The total original weight of all pieces failing the test, expressed as percentage of the total original weight of the sample, shall be considered as percentage loss by breaking.

(b) *Loss by crumbling and flaking.* After the loss by breaking has been determined, the pieces that have not failed as described in (*a*) shall be screened on the 1/2 inch sieve and the material retained on the sieve shall be weighed for purposes required in (c). The pieces which have failed by breaking shall then be added to the material remaining on the 1/2 inch sieve and screened. The difference between the total original weight of the sample and the total weight of the material retained on the 1/2 inch sieve expressed as percentage of the total original weight, shall be considered as percentage loss by crumbling and flaking.

(c) *Combined loss.* The difference between the total original weight of the sample and the weight of the material retained on the 1/2 inch sieve, after screening the pieces which did not fail by breaking, expressed as percentage of the total original weight, shall be considered as percentage combined loss.

4.5 Rejection. Material that does not meet the specified requirements shall be rejected or corrected as indicated following:

4.5.1 *Soundness.* Material that does not pass the soundness test specified in 4.4.2, shall be rejected for use where grade A is specified and the source of this material shall be abandoned for purposes of grade A material.

4.5.2 *Particle shape.* Material that does not conform to the requirements for particle shape shall be rejected and the source of material that indicates a tendency toward this defect shall be abandoned.

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4.5.3 *Size gradation.* Material that does not conform to size gradation requirements shall be rejected. Material that has been rejected for this defect may be rescreened.

4.5.4 *Extraneous matter.* Material that contains oil or grease shall be rejected. Material that contains dirt, trash, or other undesirable organic material shall be rejected until acceptably corrected by washing, screening, or other appropriate means.

4.6 **Report.** A report on testing of the material shall be made for the sanitary engineer. Soundness data shall be recorded in the form shown as table I in ASTM C 88. The sieve analysis shall be reported in tabular form as follows:

Sieve analysis

Sieve size inches	Percentage retained by weight as received	Number of pieces used in soundness test	Percentage retained by weight	
			Before soundness test	After soundness test

5. PREPARATION FOR DELIVERY

5.1 The product shall be prepared for shipment and loaded on rail, truck, or other approved means of transport, to insure safe delivery to destination in accordance with the rules and regulations applicable to the mode of transportation.

6. NOTES

6.1 **Intended use.**

6.1.1 Types I, II, and III materials are used in the sanitary treatment of sewage in trickling filter systems.

6.1.2 Grade A material is intended for use where freezing and thawing temperatures are likely to occur in the filtering system.

6.1.3 Grade B material is intended for use where freezing temperature is not likely to occur in the filtering system.

6.2 **Ordering data.** Purchasers should select the desired options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, and grade of material required (1.2).

(*c*) Size of material (1.3).

(*d*) Size of lot, if different from 4.2.1.

(*e*) Size of sample required if different from (4.2.2).

6.3 **Grades.** In cases where the cost of grade A material is appreciably higher than grade B, it is recommended that the contract require grade A material for a specified depth below surface and grade B material be permitted below this depth. Grade B material should not be considered inferior to grade A, except resistance to freezing and thawing temperatures. The same degree of care should be used in its selection.

6.4 **Uniformity in size.** Uniformity in size of particles of the media is a very important factor affecting the operation of a trickling filter. For the purpose of minimizing the chances of becoming segregated during placement in the filter bed, the nominal sizes with 1 inch spread between maximum and minimum indicated in 1.3 are preferable to the sizes with 1-1/2 inch spread, if the additional cost is not too great.

Some producers may not be willing to grade material with 1 inch limits except at considerable increase in price. It is accordingly recommended that the relative cost, in the particular locality, of sizes with 1 inch spread and 1-1/2 inch spread be investigated before specifying the nominal size.

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